

## **Sean Kerr Micro-Grid PV+BESS System CALCULATIONS**

### **Array Sizing Calculation:**

Module output = 335 W

Instantaneous output:

$$P_{out} = (P_{Module})(n_{module})(n_{string}) \cdot PCE$$

$$P_{out} = (335W)(15)(8) \cdot PCE \rightarrow P_{out} = 40.2kW(80\%) \rightarrow P_{out} = 32.16kW$$

Critical Month Daily Load Support (assuming 3 sun hours of insolation in December):

$$32.16kW(3) = 96.48 kWh$$

**Coldest Day Max VOC (based on SunPower 335W module temperature coefficient and -40 degree coldest day):**

$$Coldest Day Max V_{oc} = n_{modules} \cdot v_{oc} [1 + (\Delta^{\circ}C \cdot Voc Temp Coefficient)]$$

$$Coldest Day Max V_{oc} = 15 \cdot 44.1V [1 + ((25^{\circ}C - (-40^{\circ}C)) \cdot 0.0029)]$$

$$Coldest Day Max V_{oc} = 786.2 V$$

**Battery Capacity Calculation (inverters have 96% efficiency and assuming transmission/wiring losses at 5%. DoD for Li+ assumed at 85% to be conservative and account for ageing//temperature variations):**

$$\eta = 0.96 \cdot 0.95 = 0.912 \rightarrow \eta = 91\%$$

6 x Bluetti Energy Pty Ltd 7\*B700 51.6 kWh @ 500V (with a rated discharge capacity of 50 kWh):

$$C_{actual} = C \cdot DoD \cdot \eta \rightarrow C_{actual} = (51.6kWh)(0.85)(0.91) = 39.91kWh$$

$$C_{total} = C \cdot n_{batteries} \rightarrow C_{total} = 39.91kWh(6) \rightarrow C_{total} = 239 kWh$$

$$\frac{239kWh}{100kWh} = 2.39 days of full load autonomy$$

### **Ampacity Sample Calculation:**

$$String ampacity = I_{sc\_string} \cdot 1.25^2 \rightarrow 9.92A \cdot 1.562 = 15.5A$$

Fuse sized to 15 A and conductor gauge to AWG 12 for (20 A @ 60°C)

**\*This was done for all conductors either based on max output currents (for inverter AC outputs) or max current ratings (for battery conductors).**